

Cal Cold-Read Criteria — Multi-Phase Quiver v0.2 (Lyra, pending)

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Cal Prep — Multi-Phase Quiver v0.2 cold-read criteria

Purpose

This document fixes my cold-read criteria before Lyra's v0.2 lands. Writing criteria after seeing the work risks calibrating to fit it. Per Cal sunrise discipline: "Write the referee's objection before reading the team's answer. See whether the team's work addresses it."

Lyra has identified A_{sub} (substrate operator algebra) as having multi-phase quiver structure, with Hall algebra (Ringel-Green) connection and Kac-Moody identification anticipated in v0.2+. My job: read v0.2 as if I've never seen the substrate framing, ask whether the identification holds on its own representation-theoretic terms.

Standard background I'm calibrating against

These are the external mathematical objects Lyra's v0.2 will invoke. Brief mental refresher (outside-voice content, not BST-specific):

Quivers and path algebras

- Quiver Q : directed graph with vertices Q_0 and arrows Q_1 . Relations are formal sums of paths.
- Path algebra kQ : associative algebra over field k , basis = paths (including trivial paths e_i at each vertex), multiplication = concatenation (or zero if paths don't compose).
- Bound quiver (Q, I) : quiver with two-sided ideal I of relations; bound path algebra kQ/I .
- Representations of Q : assignment of vector space V_i to each vertex + linear map $V_i \rightarrow V_j$ for each arrow $i \rightarrow j$. Forms abelian category equivalent to $\text{mod-}kQ$.

Auslander-Reiten theory

- AR-quiver: combinatorial structure of indecomposable representations + irreducible morphisms between them.
- AR-translation τ : maps non-projective indecomposable to non-injective indecomposable; periodic for finite type.
- AR-sequence (almost split sequence): short exact sequence $0 \rightarrow \tau M \rightarrow E \rightarrow M \rightarrow 0$ capturing local structure.
- Finite/tame/wild trichotomy (Drozd): a finite-dimensional algebra is exactly one of these. For path algebras kQ (Q acyclic): finite type $\Leftrightarrow Q$ underlying Dynkin ($A_n, D_n, E_6/E_7/E_8$); tame $\Leftrightarrow Q$ underlying extended Dynkin ($\tilde{A}_n, \tilde{D}_n, \tilde{E}_6/\tilde{E}_7/\tilde{E}_8$); else wild.

Hall algebras (Ringel-Green)

- Ringel's theorem: for finite-type quiver Q over F_q , the Hall algebra $H(Q)$ is isomorphic to the positive part $U_q(n_+)$ of the quantum group $U_q(g)$, where g is the Kac-Moody Lie algebra with Dynkin diagram =

underlying graph of Q .

- Green's coproduct: extends Ringel's result to give bialgebra structure; full quantum group $U_q(g)$ recovered as Drinfeld double.
- Beyond finite type: for affine/wild quivers, Hall algebra still defined but no longer simply identifies with $U_q(n_+)$; more complex structure (Kac's theorem on root systems).

Kac-Moody Lie algebras

- Generalized Cartan matrix A : integer matrix with $a_{ii} = 2$, $a_{ij} \leq 0$ for $i \neq j$, $a_{ij} = 0 \Leftrightarrow a_{ji} = 0$.
- Three types: finite (A positive definite — Dynkin), affine (A positive semidefinite, kernel 1-dim — extended Dynkin), indefinite/wild.
- Root system structure: real roots + imaginary roots; for finite type all roots real; for affine type imaginary roots $= \delta + n\delta$; for indefinite more complex.

Z_2 -graded structure

- Z_2 -graded algebra: $A = A_0 \oplus A_1$ with multiplication respecting grading.
- Bose-Fermi correspondence: A_0 = bosonic (even); A_1 = fermionic (odd).
- Superalgebra interpretation: if Z_2 -grading comes with sign rule in commutator, get Lie superalgebra structure.

Cold-read criteria — 10 questions for v0.2

When Lyra v0.2 lands, I will check each of these BEFORE reading downstream content:

Q1 — Quiver definition: vertices, arrows, relations

- What are the vertices of Lyra's quiver? Are they K-types? Wallach labels? Bose-Fermi sublattice indices?
- What are the arrows? Are they substrate operators ($\hat{Q}$, $\hat{P}_{op}$, $\hat{T}_{tick}$, etc.)? Are they substrate transitions?
- What relations I (if any) define the bound quiver (Q, I) ? Are they substrate-forced commutators?
- Cal #29 audit: was the choice of vertices/arrows/relations free, or substrate-forced? If "free," Lyra is choosing not deriving.

Q2 — Path algebra kQ correctness

- Is multiplication well-defined (concatenation of composable paths)?
- Are trivial paths e_i at each vertex correctly identified with idempotents?
- Does sum of e_i give algebra identity?
- Are relations applied correctly (paths in $I \rightarrow 0$)?

Q3 — Quiver type classification

- What is the underlying graph (undirected) of Lyra's quiver?
- Does it match a known Dynkin diagram (finite type)? Extended Dynkin (affine type)? Neither (wild)?
- For substrate D_{IV}^5 : which Dynkin type would be substrate-natural? B_2 (rank-2 root system of $SO(5)$) is the obvious candidate. Does Lyra's quiver match B_2 underlying graph?

Q4 — Hall algebra construction (Ringel-Green)

- Does Lyra invoke Hall algebra over F_q for finite q ? Which q ?
- If finite-type quiver: does Ringel's theorem give $A_{sub} \cong U_q(n_+)$ for the specific Kac-Moody Lie algebra g ?
- If non-finite type: which extension/modification of Hall algebra construction is invoked?
- Cal #27 audit: each step (Hall algebra $\rightarrow$ Drinfeld double $\rightarrow$ Kac-Moody) — substrate-forced or conventional choice?

Q5 — Kac-Moody identification of A_{sub}

- Which Kac-Moody Lie algebra g is A_{sub} identified with?
- Finite type (Dynkin): $g \in \{A_n, B_n, C_n, D_n, E_6, E_7, E_8, F_4, G_2\}$? Most likely $B_2 = \mathfrak{so}(5)$ given substrate's $SO(5,2)$ structure.
- Affine type: $g \in \{\tilde{A}_n, \tilde{B}_n, \text{etc.}\}$?
- Indefinite type: would be surprising; substrate is “tame” in some sense.
- Cal #133 partial-tautology check: identifying A_{sub} with $U_q(B_2)^+$ via Hall algebra of B_2 quiver is somewhat tautological IF the quiver was chosen to be B_2 in the first place. Substantive content: is the quiver SUBSTRATE-FORCED to be B_2 , or chosen?

Q6 — Z_2 grading source

- Where does the Z_2 grading come from?
- Candidates: (a) substrate's Bose-Fermi sublattice partition (integer/half-integer K-types per Cal #130); (b) $\text{Pin}(2)$ Z_2 grading σ_{BF} (distinct from γ^5 chirality per Cal #136); (c) Wallach K-type parity.
- Is the Z_2 -graded quiver / path algebra / Hall algebra construction standard, or is it Lyra's innovation?
- Cal #29 audit: if innovation, is the construction substrate-forced or substrate-motivated-choice?

Q7 — Cal #122 typing for the quiver structure

Apply Type A / Type B / Type C per Cal #122: - Type A (Level 1 geometric primary): quiver structure derives from D_{IV}^5 root system / Wallach K-types / geometric primaries directly. - Type B (Level 4 algebraic primary): quiver structure is abstract algebraic object identified post-hoc with substrate operator algebra. - Type C (level-crossing operational): quiver bridges geometric structure (root system) $\leftrightarrow$ algebraic structure (path algebra) $\leftrightarrow$ operational physics (substrate processes).

Default expectation: Type C is most likely given substrate's typical level-crossing pattern. But I should not pre-commit; let v0.2 evidence decide.

Q8 — Cal #29 question-shape audit

- Does the v0.2 work answer a question whose form is substrate-derived, or whose form is “find a quiver structure matching A_{sub} ”?
- The latter is back-fit risk: ANY finitely generated algebra is some quotient of some path algebra of some quiver (by Gabriel's theorem variant). So “ A_{sub} is a path algebra” is partial tautology UNLESS the quiver is uniquely substrate-forced.
- Substantive content: WHICH quiver. WHY this one. From WHAT substrate principle.

Q9 — Cal #27 forward-derivation audit at each step

Walking through: 1. Quiver Q identified — substrate-forced? 2. Relations I identified — substrate-forced? 3. Path algebra kQ/I constructed — standard; 4. Hall algebra $H(Q,I)$ constructed — depends on bound quiver setup; standard or innovative? 5. Kac-Moody identification — Ringel's theorem if finite type; else what? 6. $A_{\text{sub}} \cong$ result — at which step does substrate enter explicitly?

Each step must be either standard math OR substrate-forced. Standard math steps are fine. Substrate-forced steps need substrate-mechanism content.

Q10 — Mode 6 multi-decomposability

- If Lyra identifies A_{sub} with $U_q(B_2)^+$ via specific quiver, is the IDENTIFICATION unique?
- Could A_{sub} equally identify with $U_q(g)^+$ for some other g ? With $H(Q')^+$ for some other quiver Q' ?
- If multiple identifications possible, which is mechanistically privileged?
- Mode 6 multi-decomposability is the discipline that prevents claiming uniqueness when alternatives exist.

Honest expectations going in

- Most likely v0.2 disposition: FRAMEWORK-PLUS per Cal #126. Quiver identification with B_2 Kac-Moody algebra is substantively reasonable given D_{IV}^5 's $SO(5,2)$ structure containing $SO(5) = Spin(5)$ with rank-2 root system B_2 . But “reasonable” is not “forced” — substrate-mechanism content needed.
- Most likely typing: Type C per Cal #122. Quiver structure naturally bridges geometric primaries (K-types in Wallach lattice) and algebraic structure (substrate operator algebra). Operational content (substrate transitions via path composition) makes it level-crossing.
- Partial-tautology risk per Cal #133: “ A_{sub} is some Hall algebra of some quiver” is partially tautological by Gabriel-type arguments. Substantive content lives in WHICH quiver + WHY.
- SVC promotion gate: substrate-mechanism for the specific quiver choice (vertices + arrows + relations) must be forward-derived. Without this, v0.2 sits at FRAMEWORK-PLUS regardless of computational verification.

What would change my disposition

Three escalators that would move v0.2 above FRAMEWORK-PLUS toward SVC:

1. Quiver uniquely substrate-forced: if Lyra shows the quiver Q is uniquely determined by D_{IV}^5 structure (e.g., vertices $\leftrightarrow$ Wallach K-types via specific construction; arrows $\leftrightarrow$ substrate operators via specific construction; no choices made), partial-tautology concern dissolves.
2. Mode 6 alternatives ruled out: if alternative quiver structures Q' producing same A_{sub} identification are explicitly enumerated and ruled out by substrate-mechanism (not “we tried them and they didn’t work”), the multi-decomposability concern dissolves.
3. Operational consequence beyond identification: if the quiver/Hall algebra identification produces NEW substrate-mechanism content (e.g., specific K-type transition rates, specific commutator structures, specific observable predictions) that don’t follow trivially from the identification itself, the work is doing substantive substrate-derivation, not just labeling.

What would lower disposition

- Free choices unflagged: if vertices/arrows/relations are chosen without substrate-derivation and not labeled as such, partial-tautology concern dominates.
- Identification asserted without proof: if “ $A_{sub} = U_q(B_2)^+$ ” is claimed without verifying via Hall algebra construction explicitly, framework-level claim only.
- Z_2 grading without substrate source: if Z_2 grading is assumed without identifying which substrate Z_2 -grading (σ_{BF} vs γ^5 vs sublattice parity) is invoked, structural confusion risk.

Cross-reference

- Cal #122: Type A/B/C tier-discipline (level-classification)
- Cal #126: FRAMEWORK-PLUS tier (checkable arithmetic/structural content; substrate-mechanism gates pending)
- Cal #27 STANDING: BST-Primary-Target Forward-Derivation Discipline (result-level audit)
- Cal #29 STANDING: Question-Shape Audit Discipline (design-level audit)
- Cal #133: Partial-tautology distinction (general arithmetic vs substrate-specific content)

Cal cadence note

This prep doc is own-cadence multi-week work per Keeper menu Item 8. Written before v0.2 lands to fix criteria. When v0.2 arrives, cold-read entry will reference back to specific Q1-Q10 outcomes.

— Cal A. Brate, 2026-05-27 Wednesday ~09:00 EDT. Prep document; criteria fixed; standing reactive for Lyra v0.2 delivery.

Addendum 2026-05-27 ~11:00 EDT — $q=2$ specialization context (post-Toy-3554/3555 emergence)

Substantive update: Lyra v0.2 + Elie Toy 3554 + Toy 3555 emerged Wednesday morning with significant context refinement for this prep doc's cold-read criteria.

Key updates from Wednesday morning

1. Elie Toy 3554 identified Cal #139 cyclotomic chain = $q=2$ specialization of standard q -integers ($[n]_q = (q^n - 1)/(q - 1)$ at $q=2 = M_n$ Mersenne).
2. Elie Toy 3555 verified $q=2$ is UNIQUE among substrate $q_p \in \{3, 5, 7, 11, 13\}$ for producing Cal #139 chain. Substrate operates at $q=2$ in q -deformation framework with uniqueness — substantive substrate-mechanism content (not just identification).
3. Lyra Macdonald reframe: 5+ parameter Macdonald hypothesis $\rightarrow$ standard 2-parameter Macdonald $P_\lambda(x; q, t)$ at substrate-natural specialization ($q=2, t=\alpha=1/N_{\max}=1/137$). Both parameters RATIFIED substrate content.
4. Calibration #31 candidate filed: Substrate-Finding-as-Standard-Math-Specialization Discovery Pattern. Cal #139 cyclotomic chain = $q=2$ specialization is canonical instance.

Updated cold-read criteria for Lyra v0.7+ explicit Macdonald computation

The original Q1-Q10 cold-read criteria still apply but with updated specialization-aware framing:

Q4 (Hall algebra Ringel-Green) UPDATE: Lyra v0.7+ should produce explicit Macdonald $P_\lambda(x; q=2, t=\alpha)$ computation, not generic Hall algebra. The specialization point is RATIFIED substrate content; cold-read should verify v0.7+ uses RATIFIED $q=2$ and $t=\alpha$ specifically (not generic q, t).

Q5 (Kac-Moody identification) UPDATE: $U_q^+(B_2\text{-affine})$ at $q=2$ specifically. Standard quantum group result at $q=2$ specialization should be cited rigorously; Cal #133 check applies (q -deformation math is general; substrate-specific content is in $q=2$ specialization choice).

Q8 (Cal #29 question-shape audit) UPDATE: now that $q=2$ specialization is RATIFIED via Toy 3555 unique- q result, the back-fit risk concern shifts from “framing chosen to make A_{sub} work” to “specialization chosen because $q=2$ was already known to work.” Updated discipline: - Verify v0.7+ derivations work at $q=2$ as a CONSEQUENCE of substrate structure, not because $q=2$ is plugged in - Lyra v0.7+ should derive specialization parameters from substrate, not assume them

Q10 (Mode 6 multi-decomposability) UPDATE: Macdonald polynomial framework at $(q=2, t=\alpha)$ is more constrained than generic Macdonald — fewer alternative parameter choices. Mode 6 concern decreases at the specialization level ($q=2, t=\alpha$ uniquely substrate-natural per Toy 3555 + T2447) compared to generic Macdonald (where many specializations possible).

Calibration #31 application

Per Calibration #31 candidate v0.1 (Cal, 2026-05-27 ~10:45 EDT): substrate-finding-as-specialization recognition is GAIN not LOSS. The multi-phase quiver framework benefits: - Inherits Ringel-Green Hall algebra theorem at $q=2$ specialization - Inherits Macdonald polynomial machinery at $(q=2, t=\alpha)$ parameter point - Inherits Kac-Moody $U_q(B_2\text{-affine})$ representation theory at $q=2$

Substrate-specific content remains: WHY substrate selects $(q=2, t=\alpha)$ specifically. Toy 3555 unique- q result provides partial answer at $q=2; t=\alpha$ connection via T2447 RIGOROUSLY CLOSED ($N_{\max} = 1/\alpha$).

Cold-read tier expectation adjustment

Original expectation (pre- $q=2$): FRAMEWORK-PLUS per Cal #126 for Lyra v0.2+ multi-phase quiver work pending substrate-forced quiver/specialization.

Updated expectation (post-Toy-3555): FRAMEWORK-PLUS with SVC-tier promotion candidate path more tractable. v0.7+ explicit Macdonald computations at ($q=2$, $t=\alpha$) using established Macdonald machinery may close substrate-mechanism content faster than original “build quiver from scratch” framing suggested. Phase 0 timeline may compress (forecast caveat per Calibration #19 STANDING).

Cross-reference

- Elie Toy 3554/3555 (Wednesday May 27 morning)
- Lyra Multi-phase quiver v0.2 $\rightarrow$ v0.6 morning sequence (Hall algebra Macdonald 2-parameter)
- Calibration_31_Substrate_Finding_as_Standard_Math_Specialization_v0_1.md (Cal, 2026-05-27 ~10:45 EDT)
- Cal #141 main entry + Cal #141 addendum + Cal #142 (Wednesday May 27 referee log)
- T2447 RIGOROUSLY CLOSED ($N_{\max} = 1/\alpha$; $t=\alpha$ specialization parameter substrate-fixed)

— Cal A. Brate, 2026-05-27 Wednesday ~11:00 EDT (addendum). $q=2$ specialization context integrated; cold-read criteria Q1-Q10 still apply with specialization-aware framing; Calibration #31 GAIN-not-LOSS framing acknowledged; tier expectation adjusted for SVC-tractability with $q=2 + t=\alpha$ specialization machinery inheritance.